HW 3

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2024-03-30

Bean

Reusing Old Code

```
beandata <- read_xlsx("C:\\Users\\lbmoy2\\Downloads\\lbmoy2_beans.xlsx")

modbean <- beandata %>%
             select(-c(Eccentricity, Solidity)) %>%
               rename("LengthMaj" = MajorAxisLength, "LengthMin" = MinorAxisLength) %>%
                 filter(Class %in% c("CALI", "HOROZ", "SEKER", "SIRA", "DERMASON")) %>%
                   mutate("AvgMaj" = mean(LengthMaj), "AvgMin" = mean(LengthMin) ) %>%
                     mutate("PerimCate" = if_else(Perimeter > 804.048, "Largest", if_else(Perimeter > 7
                       arrange(desc(Extent))
```

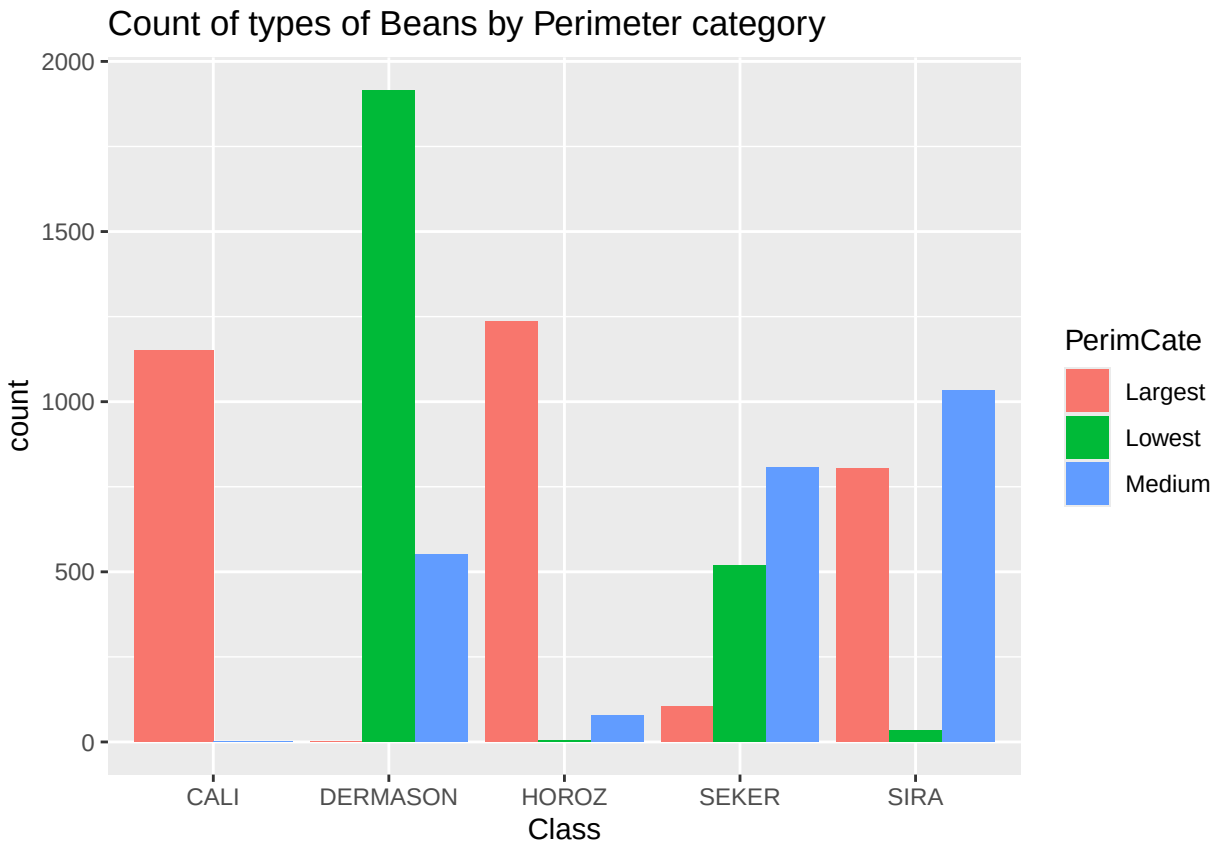
Correlation matrix

```
cor(beandata[, c("Area", "Perimeter", "AspectRatio", "ConvexArea", "EquivDiameter")], use = "complete.ob
```

```
##                    Area Perimeter AspectRatio ConvexArea EquivDiameter
## Area          1.0000000 0.9670916   0.2421310  0.9999372     0.9852003
## Perimeter     0.9670916 1.0000000   0.3854440  0.9680672     0.9913785
## AspectRatio   0.2421310 0.3854440   1.0000000  0.2437465     0.3042334
## ConvexArea    0.9999372 0.9680672   0.2437465  1.0000000     0.9854531
## EquivDiameter 0.9852003 0.9913785   0.3042334  0.9854531     1.0000000
```

GGplot bar plot

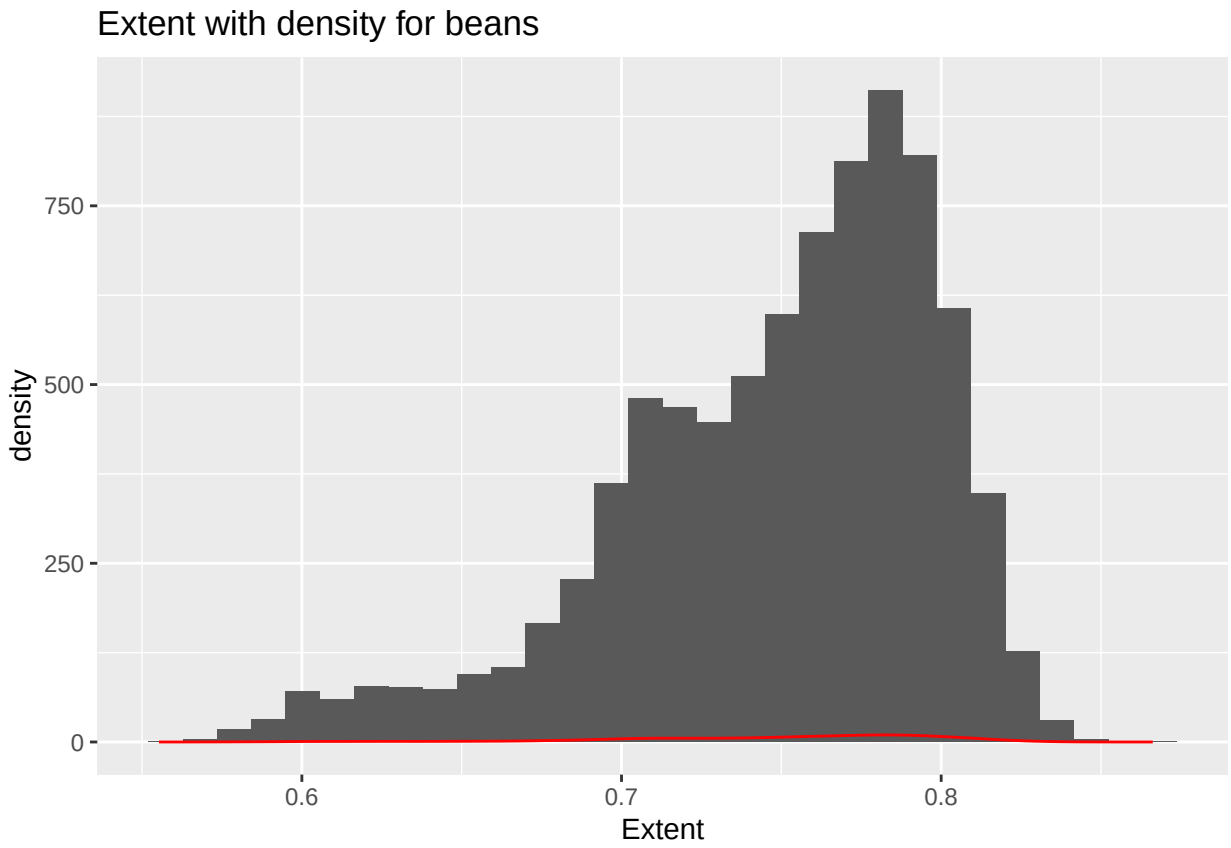
```
g <-ggplot(data = modbean, aes(x = Class, fill = PerimCate))+ labs(title = "Count of types of Beans by
g + geom_bar(position = "dodge")
```



GGplot histogram

```
h <- ggplot(data = modbean, aes(x = Extent)) + geom_histogram() + labs(title = "Extent with density for
h + geom_density(color = "red")
```

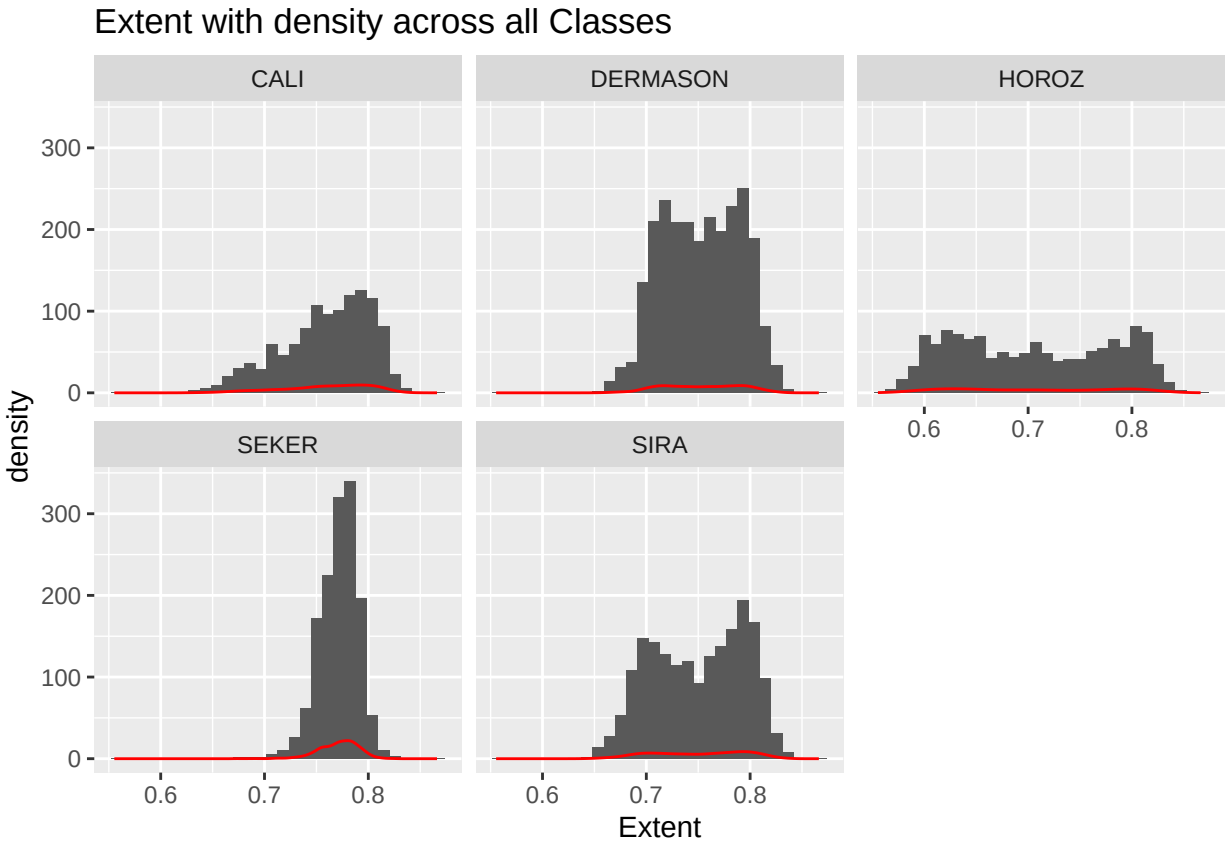
```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



Ggplot histogram for all class

```
h <- ggplot(data = modbean, aes(x = Extent)) + geom_histogram() + geom_density(color = "red") + labs(ti
h + facet_wrap(~Class)
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



Open Ended part

Reusing data that interested me URL : https://archive.ics.uci.edu/dataset/42/glass+identification

Why it interests me: I just picked this out because it said glass and I think reflections are cool. The contents don't interest me too much, nor do I know why this data is important.

```
glassdata <- read_csv("C:\\Users\\lbmoy2\\Downloads\\glass+identification\\glass.data", col_names = FALS
  rename("Id" = X1,
         "Refrac_index" = X2,
         "Sodium" = X3,
         "Magnesium" = X4,
         "Aluminum" = X5,
         "Silicon" = X6,
         "Potassium" = X7,
         "Calcium" = X8,
         "Barium" = X9,
         "Iron" = X10,
         "Type_of_glass" = X11)

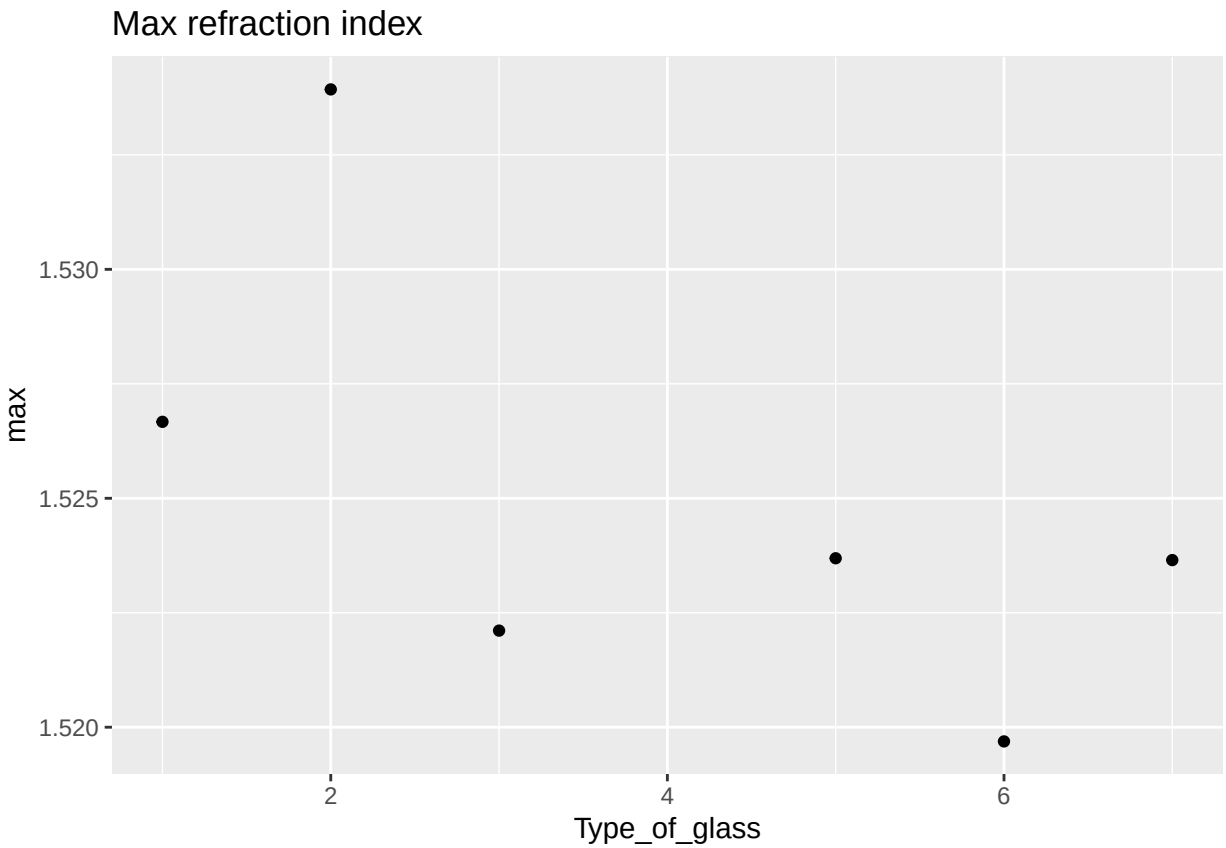
gd1 <- glassdata %>%
  group_by(Type_of_glass) %>%
    summarize( max = max(Refrac_index))
gd2 <- glassdata %>%
  summarize(smean = mean(Sodium), mmean = mean(Magnesium),amean = mean(Aluminum),simean = mean(Silicon)

modgd2 <-gd2 %>% pivot_longer(cols = 1:8, names_to = "means", values_to = "percentage")
```

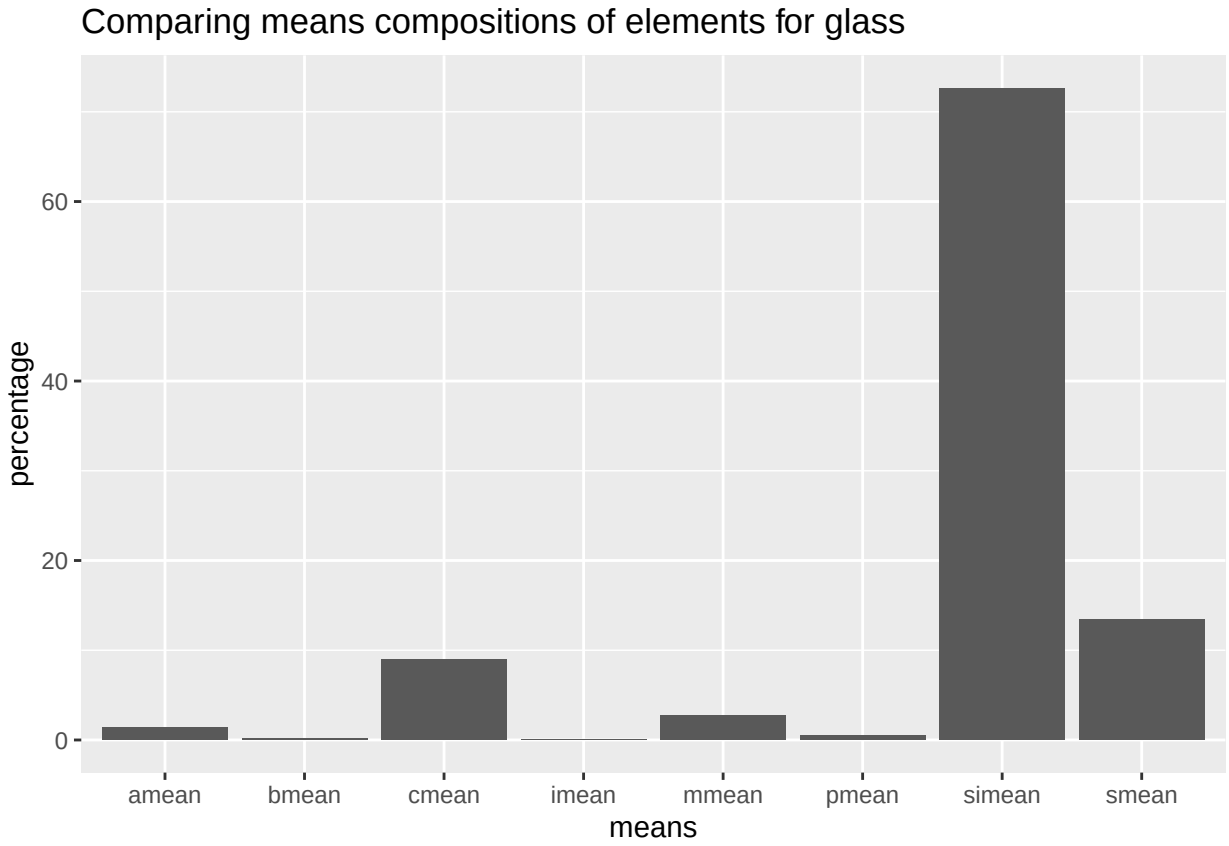
Graphing questions of interest 1) What is the max refraction index?

2) How do the measures of center of the elements compare to one another?

```
u <- ggplot(data = gd1, aes(x = Type_of_glass, y= max )) + geom_point()
u + labs(title = "Max refraction index")
```




```
a <- ggplot(data = modgd2, aes(x =means, y = percentage )) + geom_col()
a + labs(title = "Comparing means compositions of elements for glass")
```


Comparing means compositions of elements for glass
percentage
60
40
20
0
amean
bmean
cmean
imean
mmean
pmean
simean
smean
means